2.IP Addresses — Complete Guide

What is an IP Address?

When you send a letter, you need a **destination address** so the postal system knows where to deliver it. Similarly, every device on a network needs an **IP (Internet Protocol) address** — a unique identifier so data knows where to go and where it came from.

An IP address serves two purposes:

- **Identification** — Who is this device?
- **Location** — Where is this device on the network?

IPv4 — Internet Protocol Version 4

Structure

IPv4 addresses are **32-bit numbers**, written as **4 groups of decimal numbers** separated by dots. Each group is called an **octet** (8 bits).

```
192       .    168      .    1        .    1
11000000      10101000     00000001     00000001
  (8 bits)    (8 bits)     (8 bits)     (8 bits)
          = 32 bits total
```

Each octet ranges from **0 to 255** (because 8 bits = 2^8 = 256 possible values).
So IPv4 can represent **2^{32} = ~4.29 billion** unique addresses.

The Two Parts of Every IP Address

Every IPv4 address is split into two logical parts:

```
[ Network Part ]  +  [ Host Part ]
```

- **Network Part** → Identifies which network the device belongs to (like a city/street name)
- **Host Part** → Identifies the specific device within that network (like a house number)

How do we know where the network part ends and host part begins? That's where **Subnet Masks** and historically **Classful Addressing** come in.

Classful Addressing — Where It Comes From

In the **early days of the internet (1981)**, when the internet was small and managed by universities and the US military, someone had to decide how to **organize and allocate IP addresses** to different organizations.

The solution they came up with was **Classful Addressing** — dividing the entire IPv4 address space into fixed classes, each with a predetermined split between network and host bits.

The 5 Classes

```
Class A: 0.0.0.0       to  127.255.255.255
Class B: 128.0.0.0     to  191.255.255.255
Class C: 192.0.0.0     to  223.255.255.255
Class D: 224.0.0.0     to  239.255.255.255  (Multicast)
Class E: 240.0.0.0     to  255.255.255.255  (Experimental)
```

How does a router know which class an IP belongs to just by looking at it? By reading the **first few bits**:

```
Class A → starts with  0xxxxxxx  (first bit is 0)
Class B → starts with  10xxxxxx  (first two bits are 10)
Class C → starts with  110xxxxx  (first three bits are 110)
Class D → starts with  1110xxxx  (multicast)
Class E → starts with  1111xxxx  (experimental)
```

Class A – For Giant Organizations

```
First bit: 0
Network bits: 8      Host bits: 24
Format: [Network 8 bits].[Host 24 bits].[Host].[Host]

Range: 1.0.0.0  to  126.255.255.255
(0.x.x.x is reserved, 127.x.x.x is loopback)

Number of networks:  2^7 = 126 usable networks
Hosts per network:   2^24 - 2 = ~16.7 million hosts
```

- **Who got Class A?** Big corporations and governments – IBM, Apple, GE, the US Army. Each got an entire Class A block – meaning one organization could have 16 million IP addresses.
- **Default Subnet Mask:** `255.0.0.0`

Class B – For Medium-Large Organizations

```
First two bits: 10
Network bits: 16     Host bits: 16
Format: [Network].[Network].[Host].[Host]

Range: 128.0.0.0  to  191.255.255.255

Number of networks:  2^14 = 16,384 networks
Hosts per network:   2^16 - 2 = 65,534 hosts
```

- **Who got Class B?** Universities, mid-size companies, ISPs
- **Default Subnet Mask:** `255.255.0.0`

Class C – For Small Organizations

```
First three bits: 110
Network bits: 24     Host bits: 8
Format: [Network].[Network].[Network].[Host]

Range: 192.0.0.0  to  223.255.255.255

Number of networks:  2^21 = ~2 million networks
Hosts per network:   2^8 - 2 = 254 hosts
```

- **Who got Class C?** Small businesses, small offices
- **Default Subnet Mask:** `255.255.255.0`

Why -2 for Host Count?

In every network, **2 addresses are always reserved**:

- **Network Address** → All host bits = 0 (e.g., `192.168.1.0`) – identifies the network itself
- **Broadcast Address** → All host bits = 1 (e.g., `192.168.1.255`) – sends data to ALL devices on the network

So usable hosts = $2^n - 2$, where n = number of host bits.

The Problem With Classful Addressing – Why It Failed

Classful addressing had a **huge flaw**: it was extremely wasteful.
Imagine a company needs **300 IP addresses**:

- Class C gives only 254 → **not enough**
- Class B gives 65,534 → they only use 300, **65,234 addresses wasted**

This is called **IP address wastage**, and as the internet grew rapidly in the late 1980s and early 1990s, the address space was getting **exhausted very quickly**.

Also, routing tables were exploding in size because every class block had to be tracked separately.

This led to the invention of **CIDR**.

CIDR — Classless Inter-Domain Routing (1993)

CIDR threw away the fixed class boundaries and said: **"You can split the network/host boundary anywhere you want."**

Instead of relying on classes, CIDR uses a **prefix length** written with a `/` (slash) notation:

```
192.168.1.0/24
```

The `/24` means: **the first 24 bits are the network part**, the remaining 8 bits are the host part.

```
/8   →  255.0.0.0      (Class A equivalent)
/16  →  255.255.0.0    (Class B equivalent)
/24  →  255.255.255.0  (Class C equivalent)
/26  →  255.255.255.192  (custom — 64 addresses)
/30  →  255.255.255.252  (custom — 4 addresses, good for point-to-point links)
```

How Does the Subnet Mask Work?

A **subnet mask** is a 32-bit number where all network bits = `1` and all host bits = `0`.

```
IP Address:    192.168.1.10    →  11000000.10101000.00000001.00001010
Subnet Mask:   255.255.255.0  →  11111111.11111111.11111111.00000000
                                    (24 ones = /24)

AND operation gives Network Address:
               192.168.1.0    →  11000000.10101000.00000001.00000000
```

By ANDing the IP with the subnet mask, routers extract the **network address** and know which network a packet belongs to.

Subnetting — Dividing a Network into Smaller Pieces

Say your organization is given `192.168.1.0/24` — that's 254 usable hosts. But you have 4 departments and want to **isolate their traffic**. You can **subnet** it:

```
Original:    192.168.1.0/24  → 254 hosts (1 network)

Subnet into /26 (borrow 2 bits from host):
  Subnet 1:  192.168.1.0/26    → hosts: .1 to .62    (62 hosts) — HR
  Subnet 2:  192.168.1.64/26   → hosts: .65 to .126  (62 hosts) — Engineering
  Subnet 3:  192.168.1.128/26  → hosts: .129 to .190 (62 hosts) — Finance
  Subnet 4:  192.168.1.192/26  → hosts: .193 to .254 (62 hosts) — Management
```

Subnetting gives you **security, traffic isolation, and efficient address use**.

Types of IP Addresses

1. Public IP Address

- **Visible on the internet**
- Assigned by your ISP
- **Globally unique** — no two devices on the internet share the same public IP
- Used when you browse the internet, host a server, etc.
- Example: `142.250.195.46` (a Google IP)

2. Private IP Address

- Used **within a private network** (home, office)
- **Not routable on the internet** — the internet doesn't know about them
- Multiple organizations can reuse the same private IPs — that's fine because they never appear on the public internet
- Defined by **RFC 1918**:

```
Class A private:  10.0.0.0     to  10.255.255.255    (/8)
Class B private:  172.16.0.0   to  172.31.255.255    (/12)
Class C private:  192.168.0.0  to  192.168.255.255   (/16)
```

Your home WiFi router gives your phone `192.168.1.x` — that's a private IP. Your ISP gives your router a public IP that the rest of the internet sees.

How Private + Public Work Together — NAT

NAT (Network Address Translation) is the mechanism that lets many devices with private IPs share one public IP:

```
Your Phone    192.168.1.2  ──┐
Your Laptop   192.168.1.3  ──┼──► Router (NAT) ──► Public IP: 103.21.45.6 ──► Internet
Your TV       192.168.1.4  ──┘
```

The router keeps a **translation table**, replacing private IPs with the public IP when sending packets out, and reversing it when receiving responses. This is why IPv4 hasn't completely run out yet — NAT allows billions of devices to share a much smaller pool of public IPs.

3. Static IP Address

- **Never changes** — manually assigned and fixed
- Used for: **servers, databases, printers, network infrastructure**
- If you're hosting a website, your server needs a static IP so the DNS record always points to the right place
- More expensive (ISPs charge extra)

4. Dynamic IP Address

- **Changes periodically** — assigned automatically by **DHCP (Dynamic Host Configuration Protocol)**
- Used for: **regular user devices** — laptops, phones, home internet connections
- When your phone connects to WiFi, the router's DHCP server assigns it a free IP from a pool
- When you disconnect, that IP is returned to the pool for others to use
- Cost-effective — ISPs can serve many customers with fewer IPs since not everyone is online simultaneously

5. Loopback Address

- `127.0.0.1` (also called **localhost**)
- Used by a device to **talk to itself**
- Never leaves the machine — traffic sent to `127.0.0.1` is looped back internally
- Used by developers to test servers locally (`http://localhost:3000`)
- The entire `127.0.0.0/8` range is reserved for loopback
-

6. APIPA — Automatic Private IP Addressing

- Range: `169.254.0.0/16`
- When a device **can't reach a DHCP server**, it assigns itself an APIPA address automatically
- Only works for **local link communication** — can't reach the internet
- If you see a `169.254.x.x` address, it usually means **DHCP failed** — a sign of network trouble

7. Multicast Address

- Range: `224.0.0.0` to `239.255.255.255` (Class D)
- Sends data to **a group of interested receivers** simultaneously
- Unlike **broadcast** (sends to everyone) or **unicast** (sends to one), multicast sends to **subscribed members only**
- Used in: video streaming, online gaming, stock market data feeds, IPTV

8. Broadcast Address

- Sends data to **all devices** on a network simultaneously
- **Limited broadcast**: `255.255.255.255` — reaches all hosts on the local network
- **Directed broadcast**: e.g., `192.168.1.255` — reaches all hosts in `192.168.1.0/24`

- Used by DHCP (your device broadcasts "anyone have an IP for me?" when it joins a network)

IPv6 — The Solution to IPv4 Exhaustion

By the early 1990s it was clear: **4.29 billion IPv4 addresses would not be enough** for the growing internet. IPv6 was designed in **1998** to solve this permanently.

Structure

IPv6 addresses are **128 bits** long, written as **8 groups of 4 hexadecimal digits** separated by colons:

```
2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Simplification Rules

IPv6 has rules to shorten addresses:
Rule 1: Drop leading zeros in each group:

```
2001:0db8  →  2001:db8
```

Rule 2: Replace one or more consecutive all-zero groups with `::` (only once):

```
2001:db8:0:0:0:0:0:1  →  2001:db8::1
```

How Many Addresses Does IPv6 Offer?

```
2^128 = 340,282,366,920,938,463,463,374,607,431,768,211,456

= ~340 undecillion addresses
= 340 trillion trillion trillion addresses
```

That's enough to give **every grain of sand on Earth its own IP address** — billions of times over. We will never run out.

IPv4 vs IPv6 — Key Differences

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Total addresses	~4.3 billion	~340 undecillion
Notation	Decimal, dots	Hexadecimal, colons
Header size	20 bytes (variable)	40 bytes (fixed)
NAT required	Yes (due to scarcity)	No (abundance)
IPSec (security)	Optional	Built-in
Broadcast	Yes	No (uses Multicast)
DHCP	Needed	Optional (SLAAC — self-configuring)
Loopback	127.0.0.1	::1